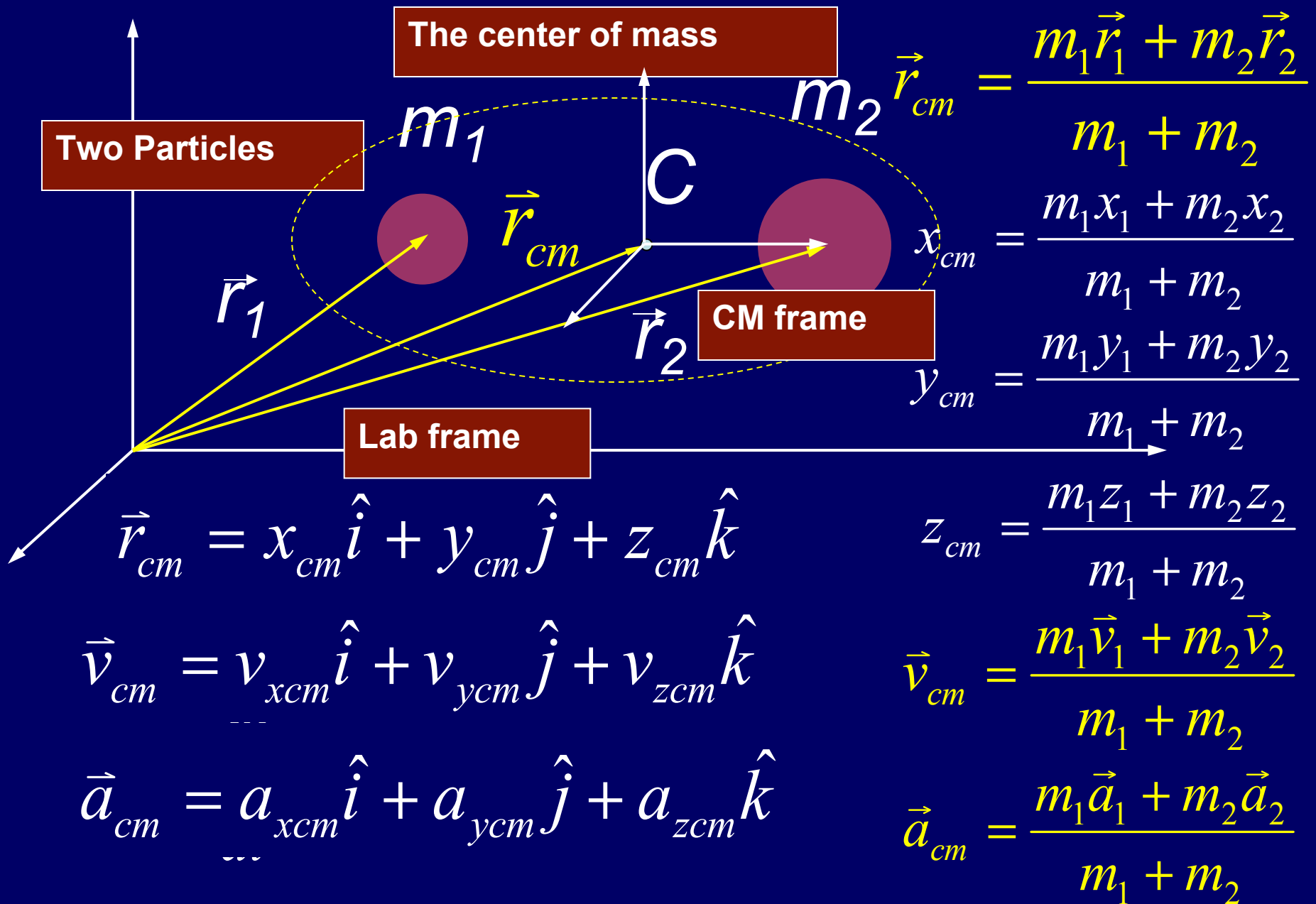


Chapter 7

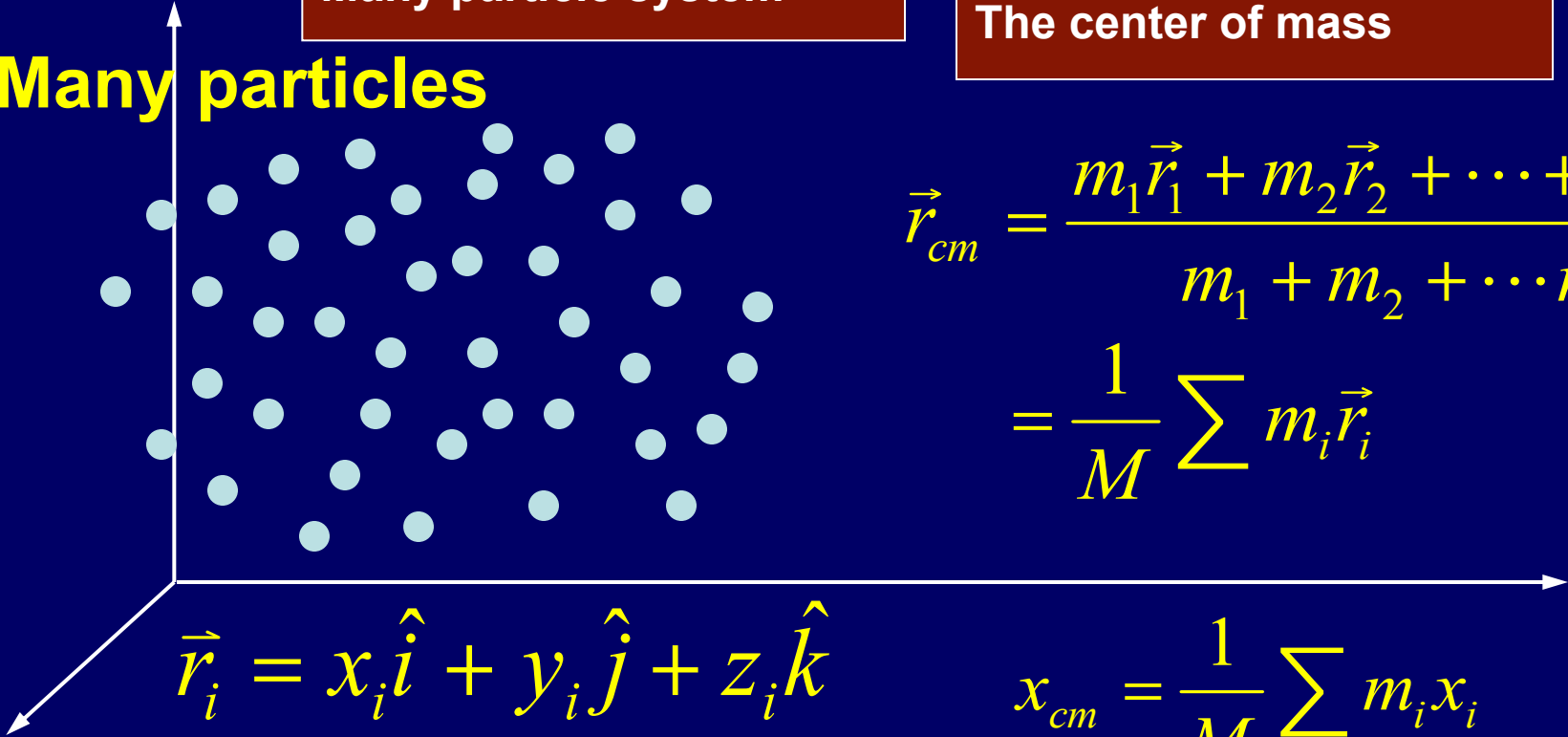
Systems of Particles



Many particle system

The center of mass

Many particles



$$\begin{aligned}\vec{r}_{cm} &= \frac{m_1\vec{r}_1 + m_2\vec{r}_2 + \cdots + m_N\vec{r}_N}{m_1 + m_2 + \cdots + m_N} \\ &= \frac{1}{M} \sum m_i \vec{r}_i\end{aligned}$$

$$\vec{r}_i = x_i \hat{i} + y_i \hat{j} + z_i \hat{k}$$

$$M = m_1 + m_2 + \cdots + m_N = \sum m_n.$$

$$\vec{v}_{cm} = \frac{1}{M} \sum m_i \vec{v}_i$$

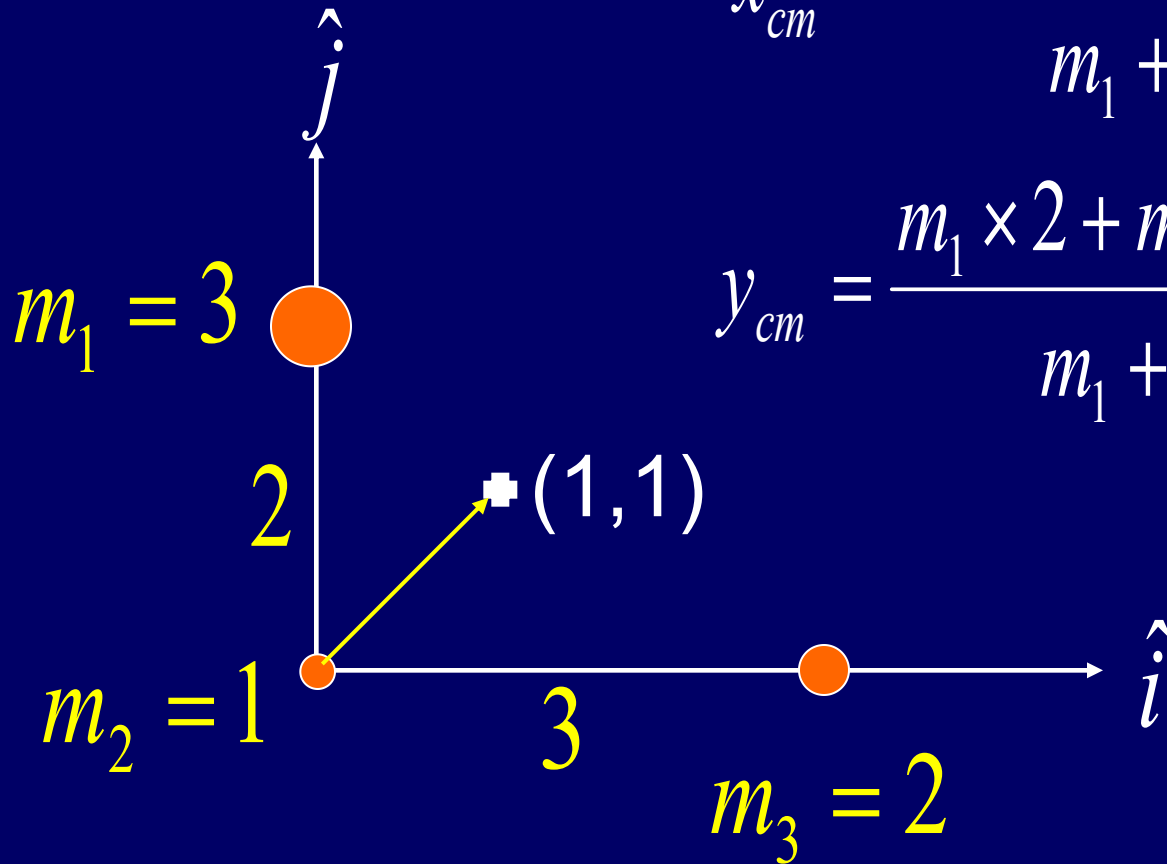
$$\vec{a}_{cm} = \frac{1}{M} \sum m_i \vec{a}_i$$

$$x_{cm} = \frac{1}{M} \sum m_i x_i$$

$$y_{cm} = \frac{1}{M} \sum m_i y_i$$

$$z_{cm} = \frac{1}{M} \sum m_i z_i$$

Example



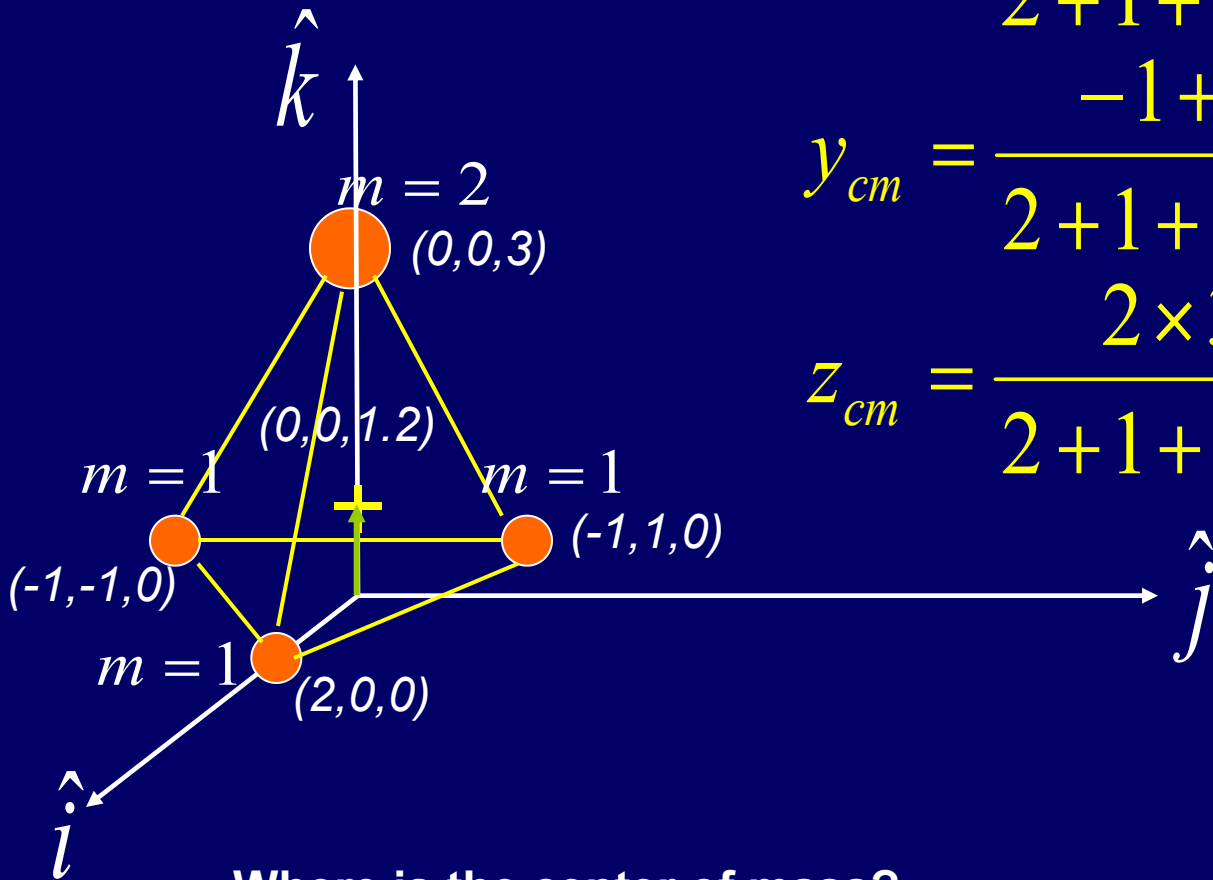
$$x_{cm} = \frac{m_1 \times 0 + m_2 \times 0 + m_3 \times 3}{m_1 + m_2 + m_3} = \frac{6}{6} = 1$$

$$y_{cm} = \frac{m_1 \times 2 + m_2 \times 0 + m_3 \times 0}{m_1 + m_2 + m_3} = \frac{6}{6} = 1$$

$$\vec{r}_{cm} = \hat{i} + \hat{j}$$

Where is the center of mass?

Example



Where is the center of mass?

$$x_{cm} = \frac{2-1-1}{2+1+1+1} = 0;$$

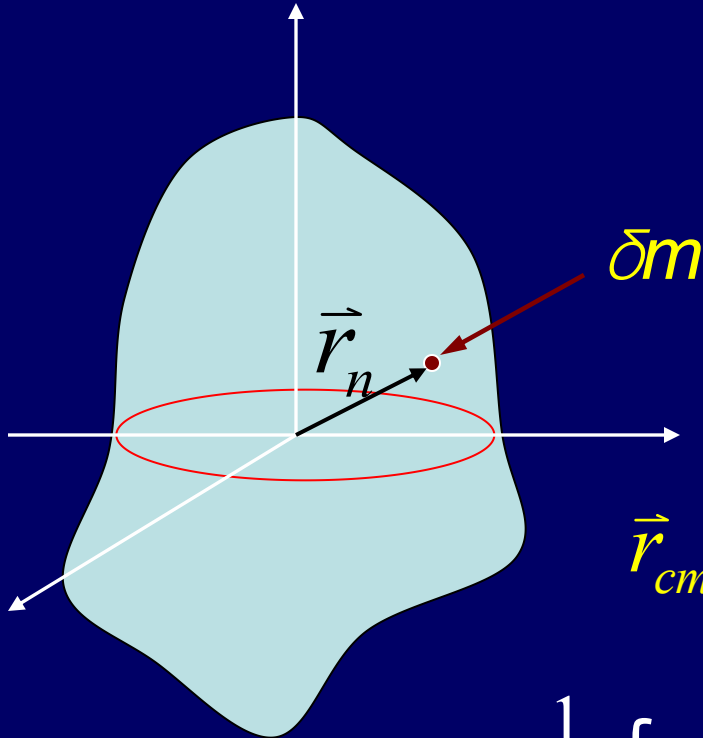
$$y_{cm} = \frac{-1+1}{2+1+1+1} = 0$$

$$z_{cm} = \frac{2 \times 3}{2+1+1+1} = 1.2$$

$$\vec{r}_{cm} = 1.2\hat{k}$$

A solid body (Many particles)

Center of mass of
solid objects



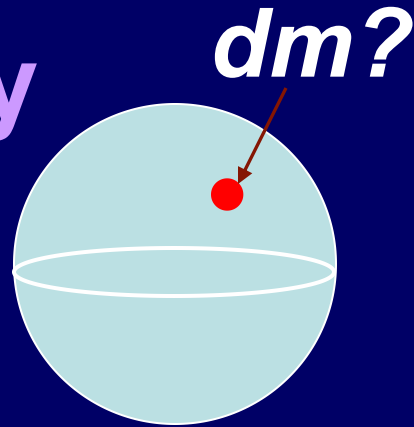
$$\vec{r}_{cm} = \frac{1}{M} \sum \vec{r}_n \delta m_n$$

$$\vec{r}_{cm} = \frac{1}{M} \lim_{\delta m \rightarrow 0} \sum \vec{r}_n \delta m_n = \frac{1}{M} \int \vec{r} dm$$

$$x_{cm} = \frac{1}{M} \int x dm, \quad y_{cm} = \frac{1}{M} \int y dm, \quad z_{cm} = \frac{1}{M} \int z dm.$$

$$dm = ?$$

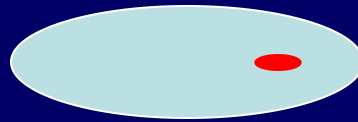
solid body



$$dm = \rho dV$$

ρ is mass density
of volume

plane



$$dm = \sigma dS, \sigma \text{ is mass density of plane}$$

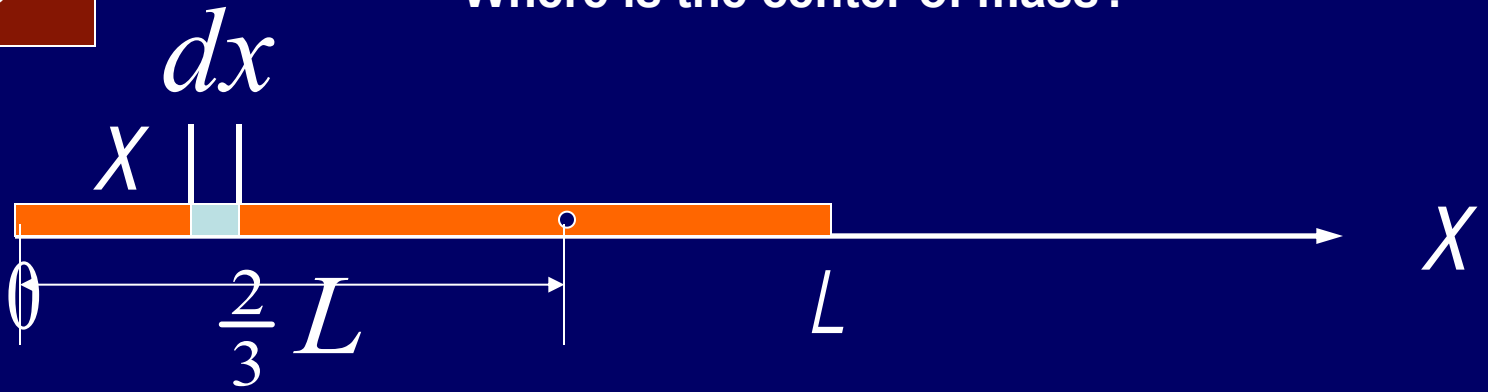
line



$$dm = \lambda dl, \lambda \text{ is mass density of line}$$

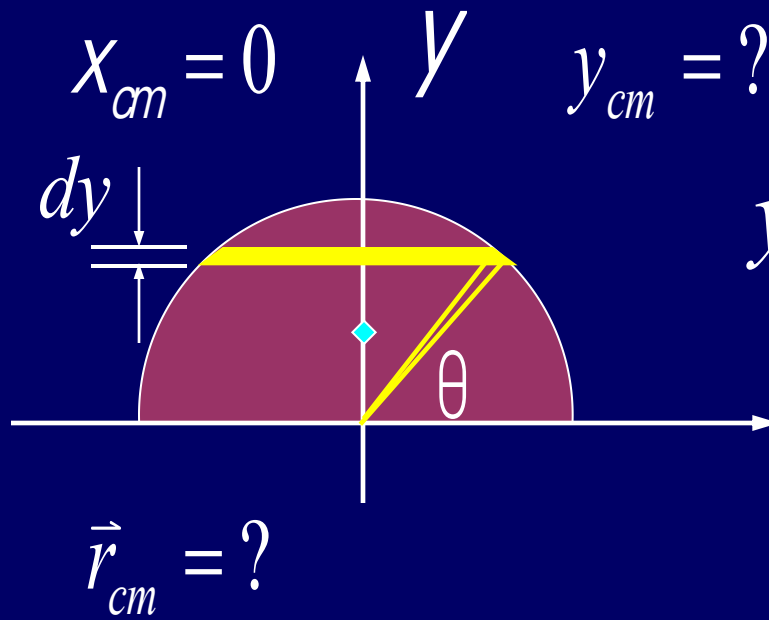
Example

Where is the center of mass?



$$\lambda = \lambda(x) = ax \quad dm = ax dx$$

$$x_{cm} = \frac{\int_0^L x dm}{\int_0^L dm} = \frac{\int_0^L ax^2 dx}{\int_0^L ax dx} = \frac{\frac{1}{3} aL^3}{\frac{1}{2} aL^2} = \frac{2}{3} L$$



$$ds = 2R \cos \theta dy$$

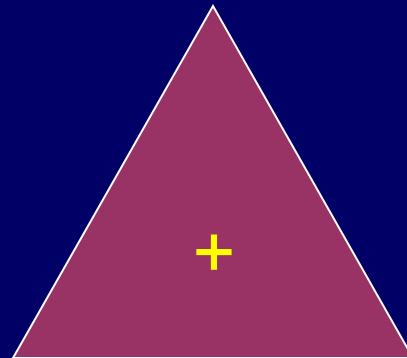
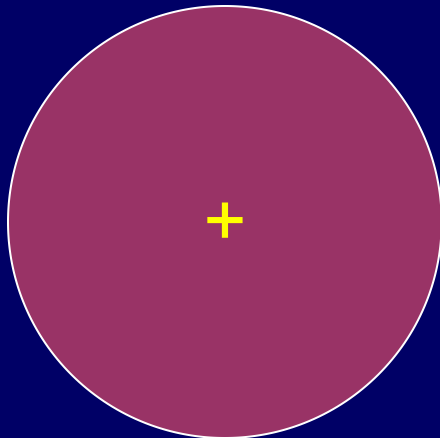
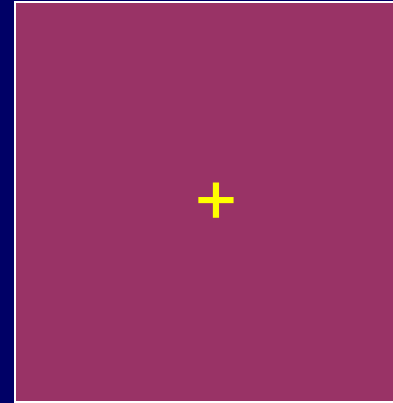
$$y = R \sin \theta \quad dy = R \cos \theta d\theta$$

$$ds = 2R^2 \cos^2 \theta d\theta$$

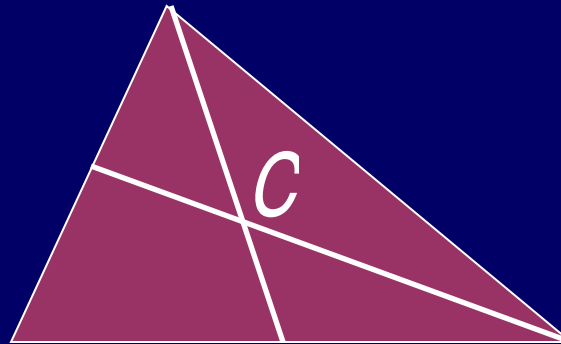
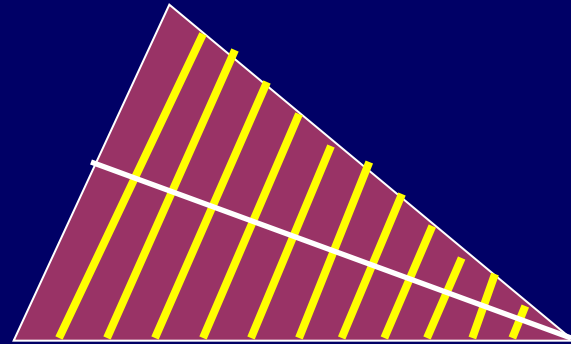
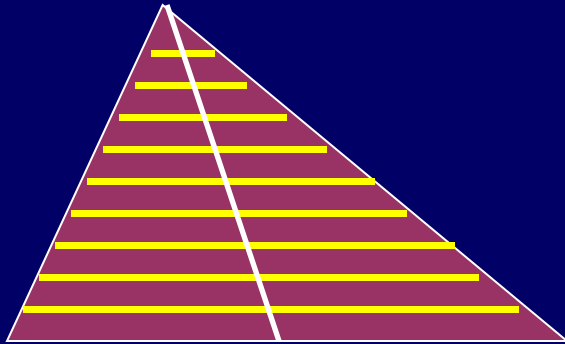
$$dm = \sigma ds = 2\sigma R^2 \cos^2 \theta d\theta$$

$$y_{cm} = \frac{\int_0^{\pi/2} y dm}{\int_0^{\pi/2} dm} = \frac{\int_0^{\pi/2} R \sin \theta \sigma 2R^2 \cos^2 \theta d\theta}{\int_0^{\pi/2} \sigma 2R^2 \cos^2 \theta d\theta} = \frac{4R}{3\pi} = 0.425R$$

Center of mass of solid objects

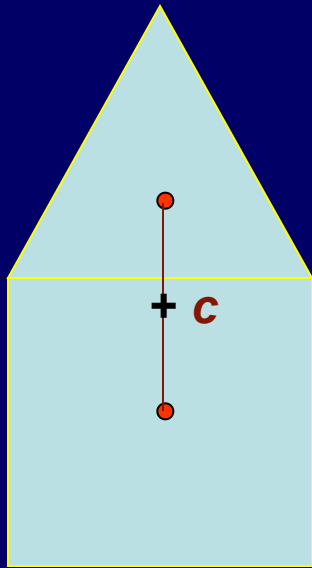


Center of mass of solid objects

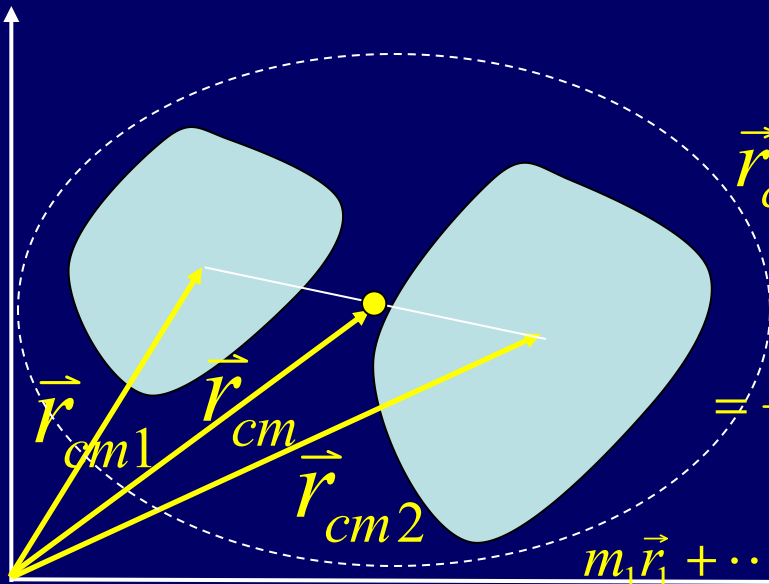


Center of mass of solid objects

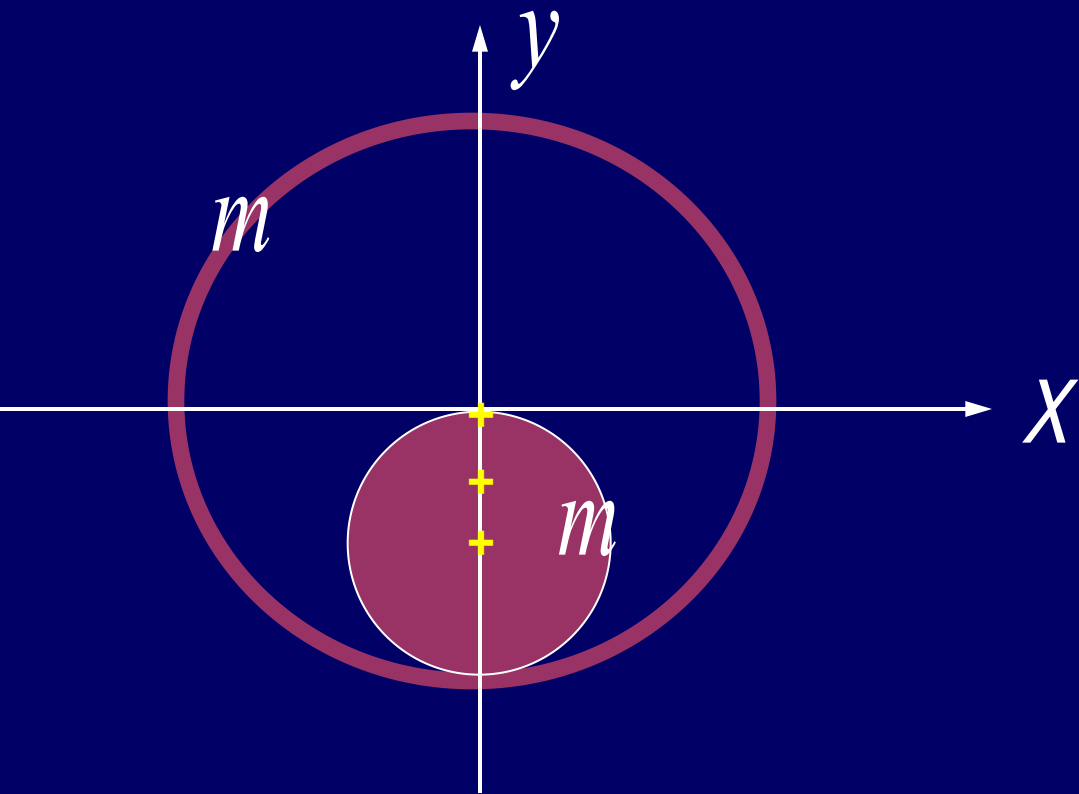
Why?



- If a irregular objects can be divided into several parts
- We can find the center of mass of each part ,
 - Treating each part as a particle located at its own center of mass
- We can find the center of mass of the combination.



$$\begin{aligned}
 \vec{r}_{cm} &= \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \cdots + m_N \vec{r}_N}{m_1 + m_2 + \cdots m_N} \\
 &= \frac{(m_1 \vec{r}_1 + \cdots + m_M \vec{r}_M) + (m_{M+1} \vec{r}_{M+1} + \cdots + m_N \vec{r}_N)}{m_1 + m_2 + \cdots m_N} \\
 &= \frac{\frac{m_1 \vec{r}_1 + \cdots + m_M \vec{r}_M}{m_1 + \cdots + m_M} (m_1 + \cdots m_M) + (m_{M+1} \vec{r}_{M+1} \cdots + m_N \vec{r}_N)}{m_1 + m_2 + \cdots m_N} \\
 &= \frac{M_1 \vec{r}_{cm1} + (m_{M+1} \vec{r}_{M+1} \cdots + m_N \vec{r}_N)}{m_1 + m_2 + \cdots m_N} \\
 &= \frac{M_1 \vec{r}_{cm1} + M_2 \vec{r}_{cm2}}{M}
 \end{aligned}$$

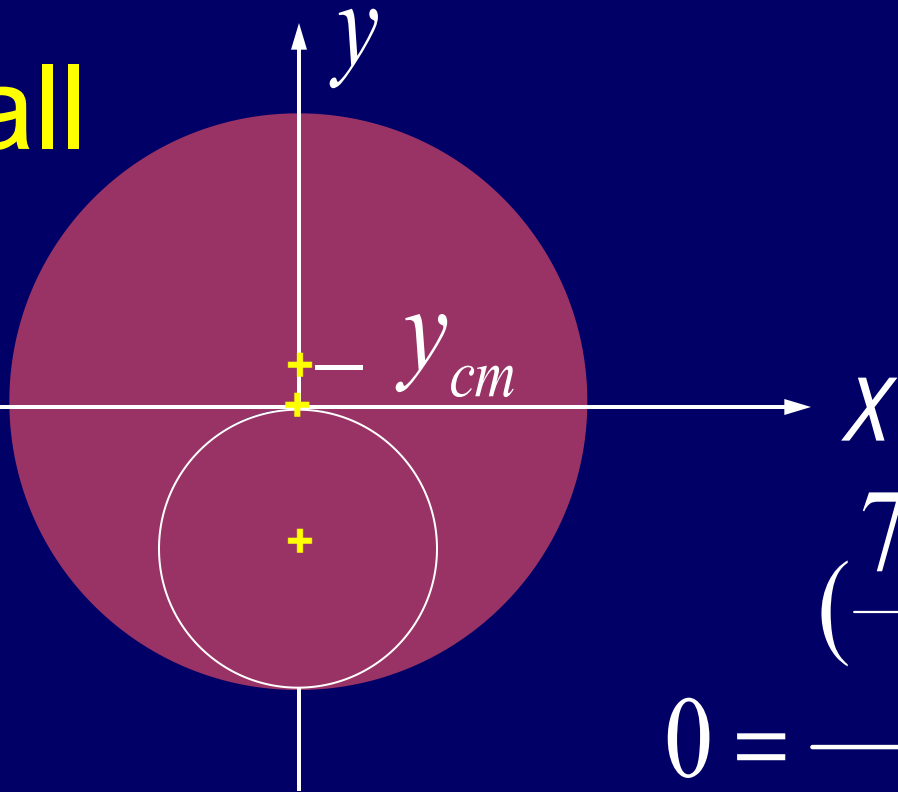


$$x_{cm} = 0$$

$$y_{cm} = \frac{m \times 0 - m \times R/2}{2m}$$

$$= -R/4$$

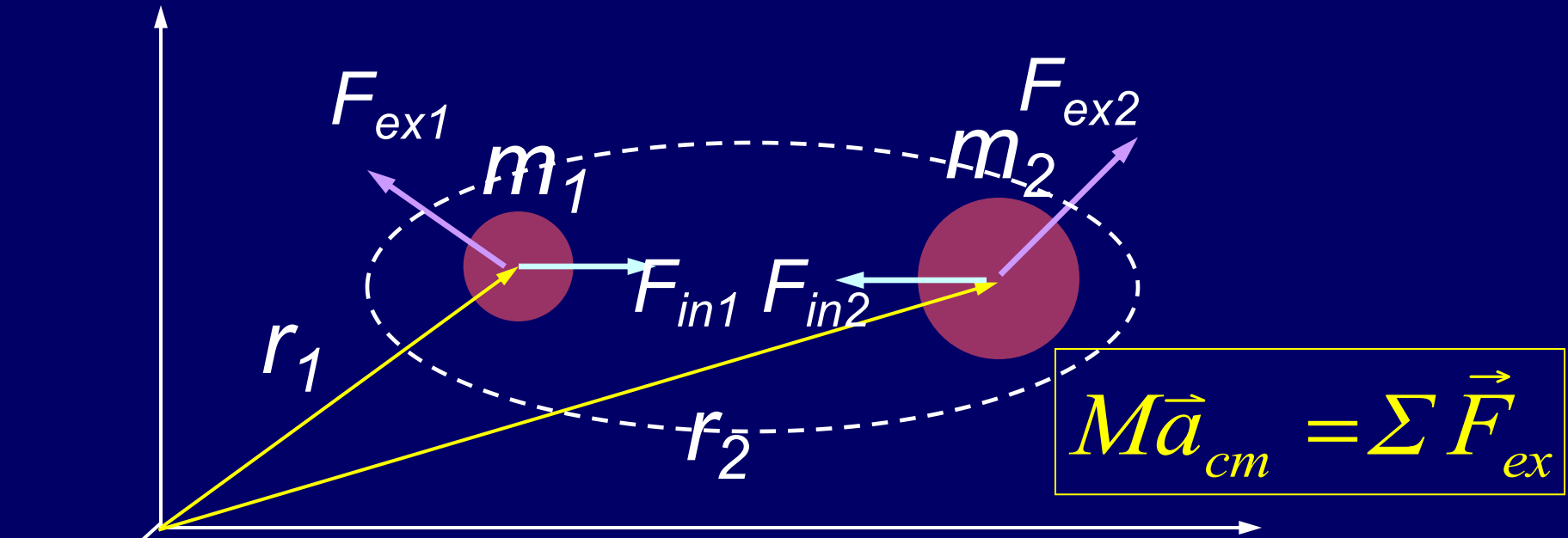
Ball



$$y_{cm} = R/14$$

$$0 = \frac{\left(\frac{7m}{8}\right)y_{cm} + \left(\frac{m}{8}\right)\left(-\frac{R}{2}\right)}{m}$$

Two-particle systems



$$m_1 \vec{a}_1 = \vec{F}_1 = \vec{F}_{ex1} + \vec{F}_{in1} \quad m_2 \vec{a}_2 = \vec{F}_2 = \vec{F}_{ex2} + \vec{F}_{in2}$$

$$\vec{a}_{cm}(m_1 + m_2) = \vec{F}_{ex1} + \vec{F}_{in1} + \vec{F}_{ex2} + \vec{F}_{in2} = \vec{F}_{ex1} + \vec{F}_{ex2} = \Sigma \vec{F}_{ex}$$

$$\vec{a}_{cm} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2}{(m_1 + m_2)}$$

Many particles

$$\sum \vec{F}_{ex} = \sum \vec{F}_{ex} + \sum \vec{F}_{in} = \sum \vec{F}_n = \sum m_n \vec{a}_n = M \vec{a}_{cm}$$

$$\sum \vec{F}_{ex} = M \vec{a}_{cm}$$

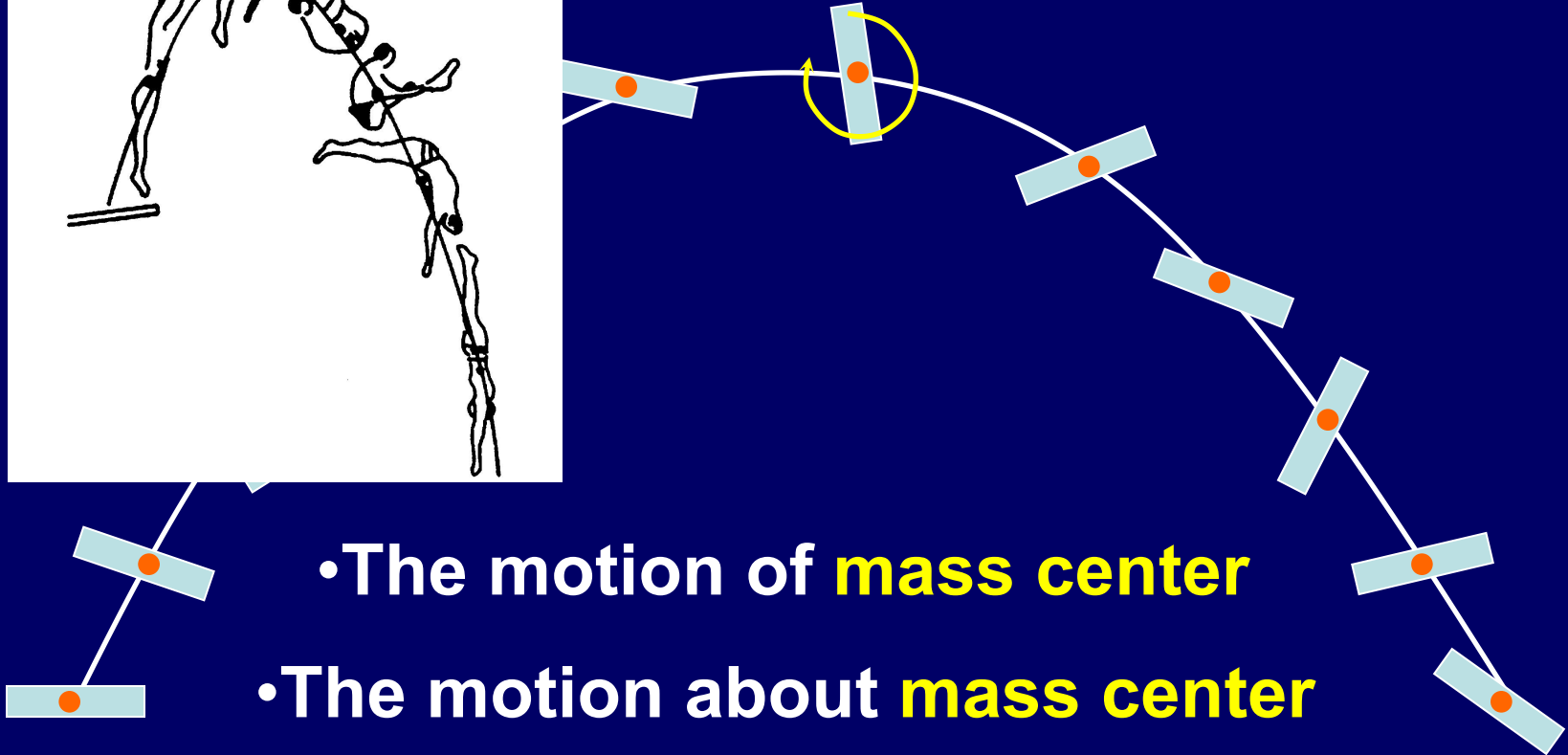
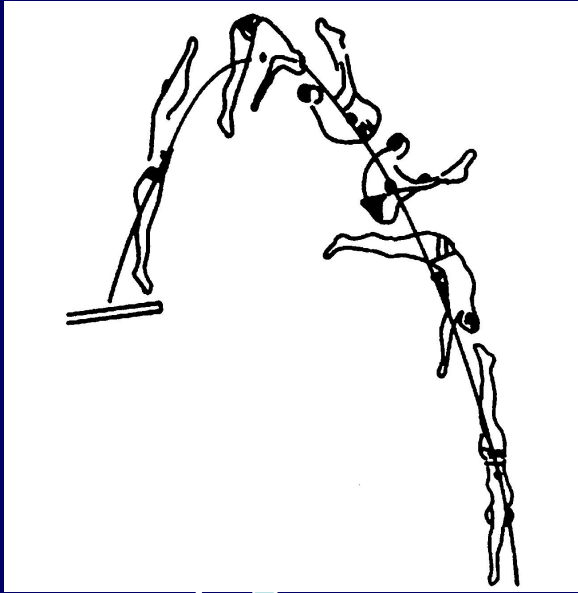
$$\sum F_{exx} = M a_{cmx}$$

$$\sum F_{exy} = M a_{cmy}$$

$$\sum F_{exz} = M a_{cmz}$$

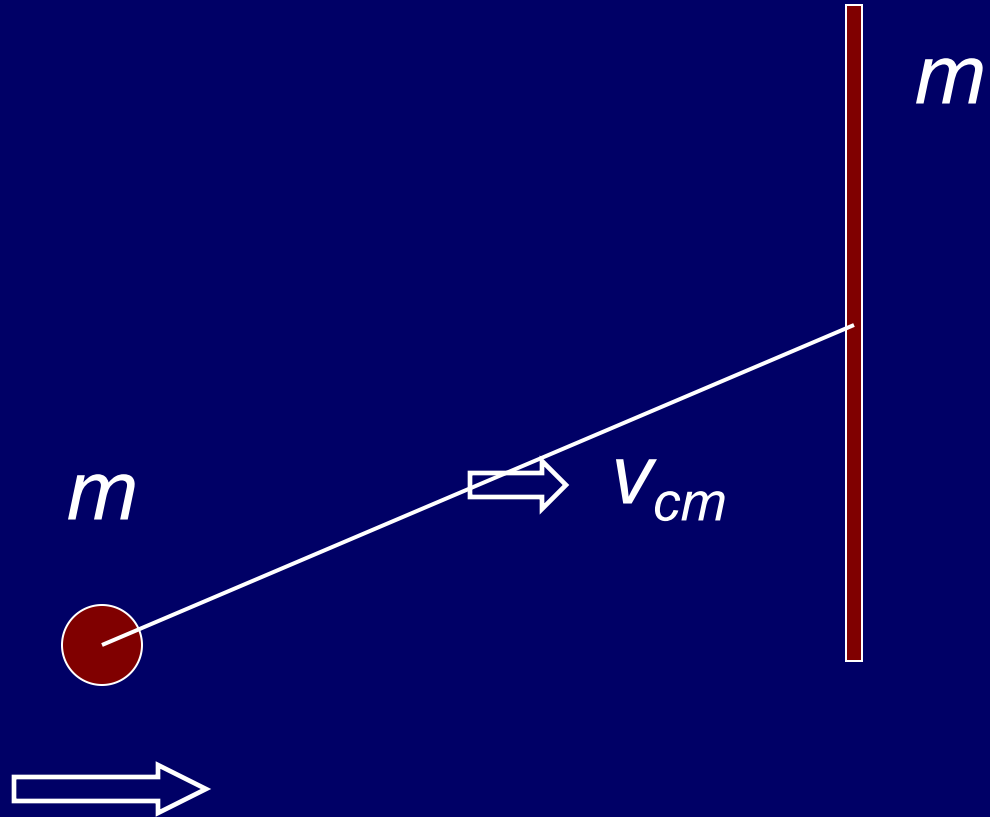
Motion of solid body

$$M\vec{a}_{cm} = \Sigma \vec{F}_{ex}$$

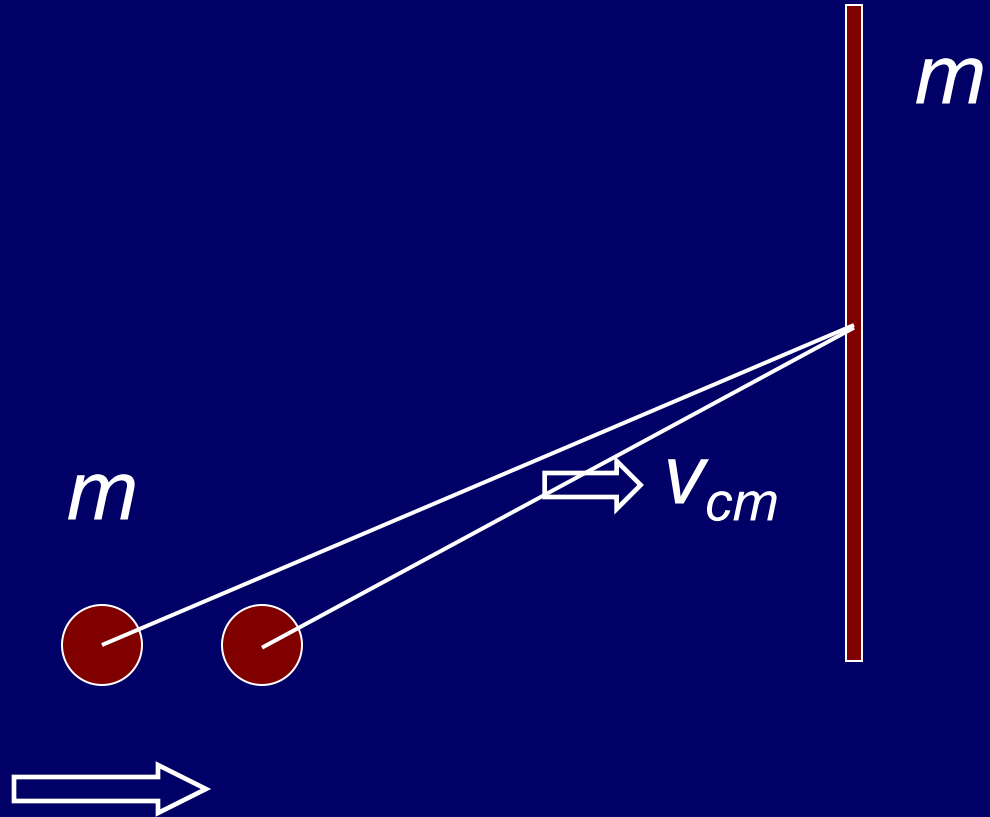


- The motion of **mass center**
- The motion about **mass center**

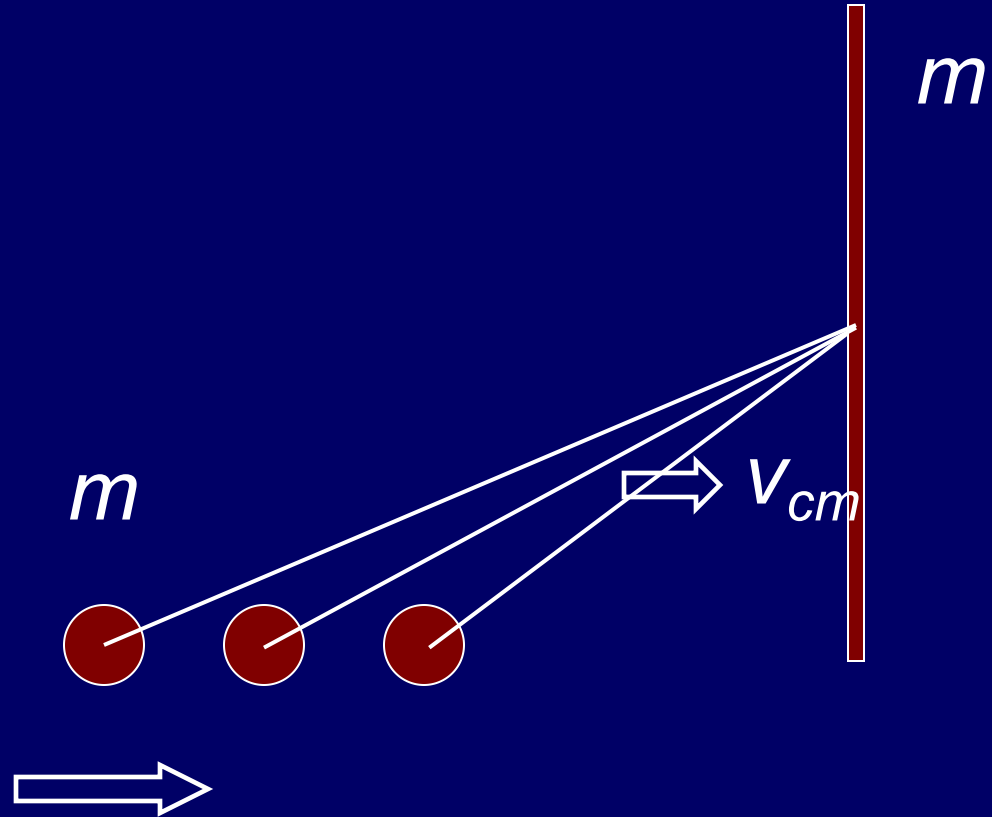
Example



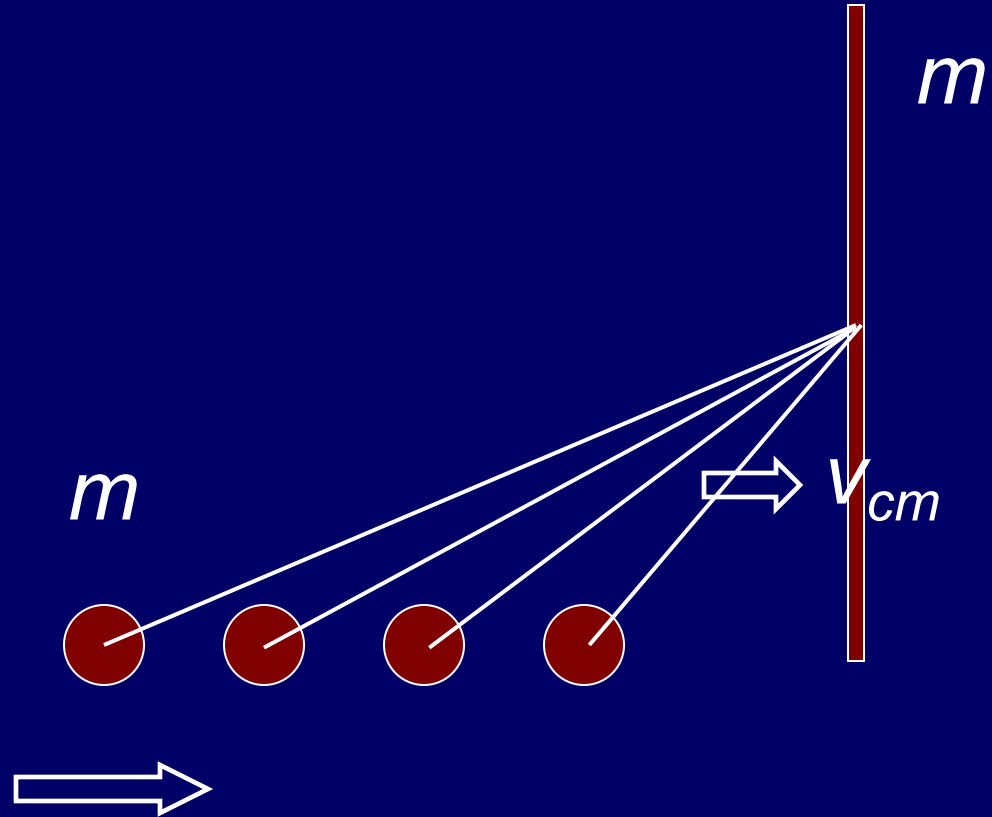
Example



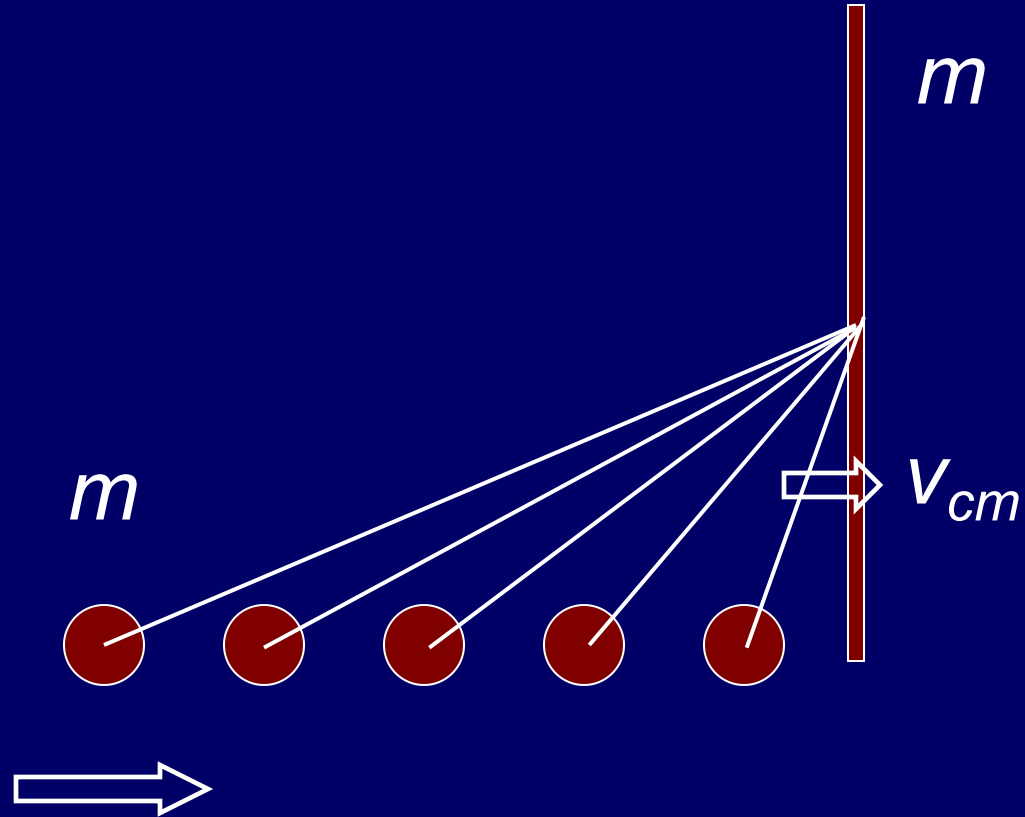
Example



Example



Example

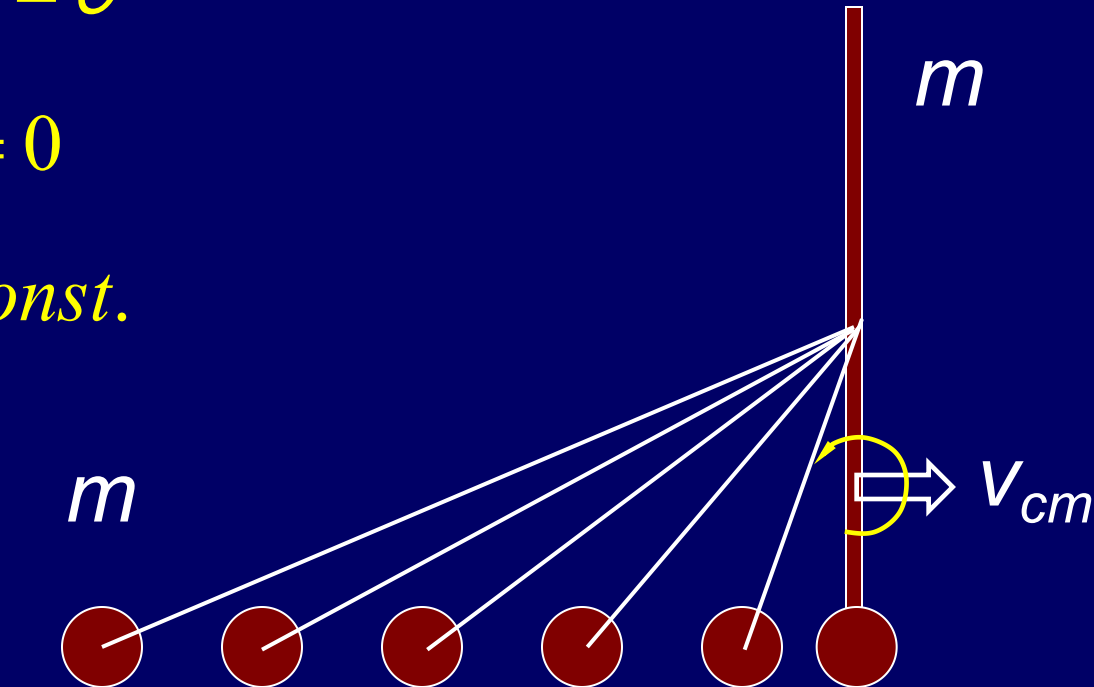


Example

$$\sum \vec{F}_{ext} = 0$$

$$\vec{a}_{cm} = 0$$

$$\vec{v}_{cm} = \text{const.}$$



- The motion of **mass center**
- The motion about **mass center**

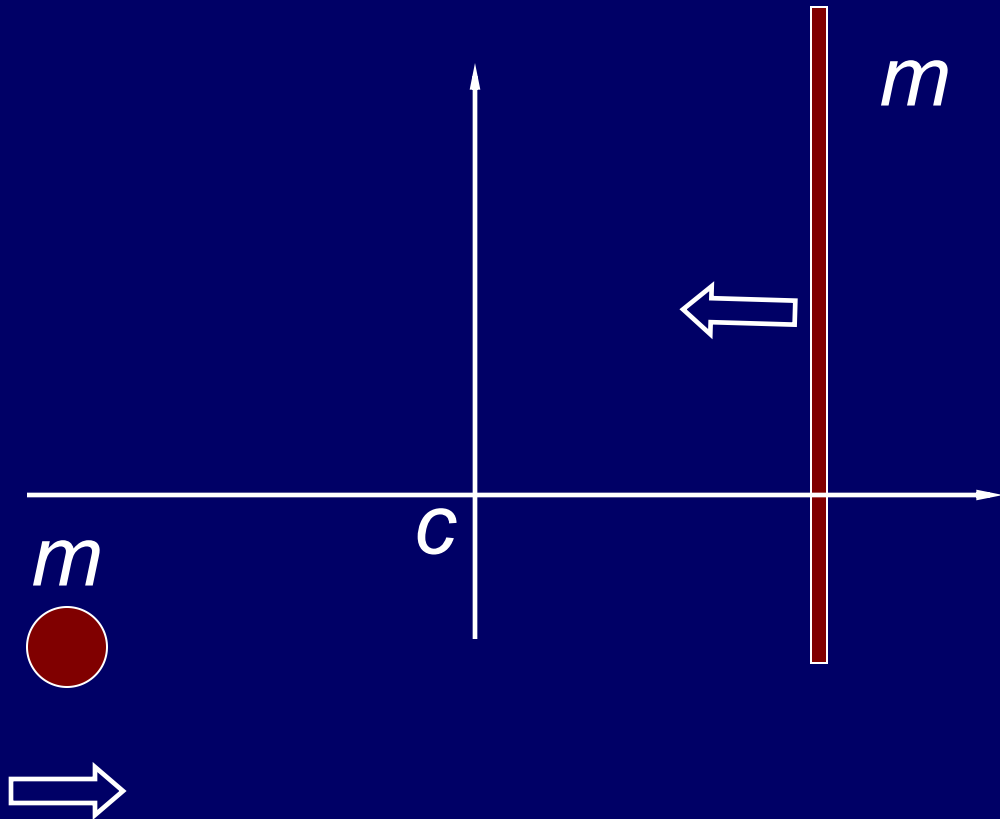
cm Frame

m

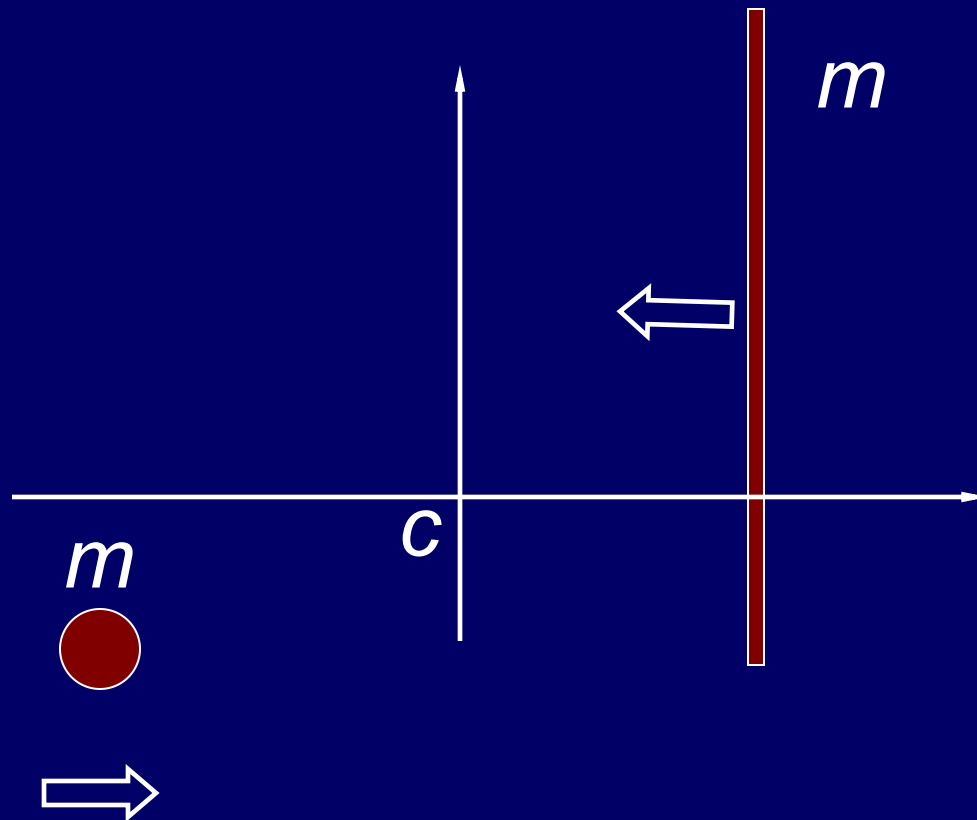


m

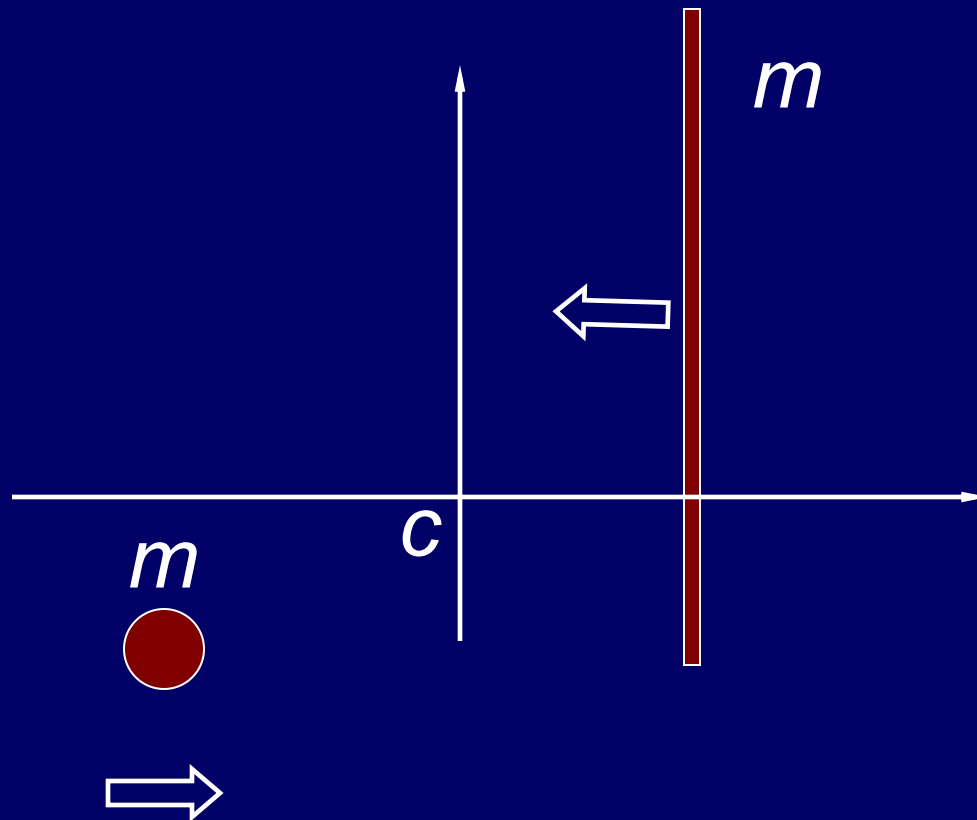
cm Frame



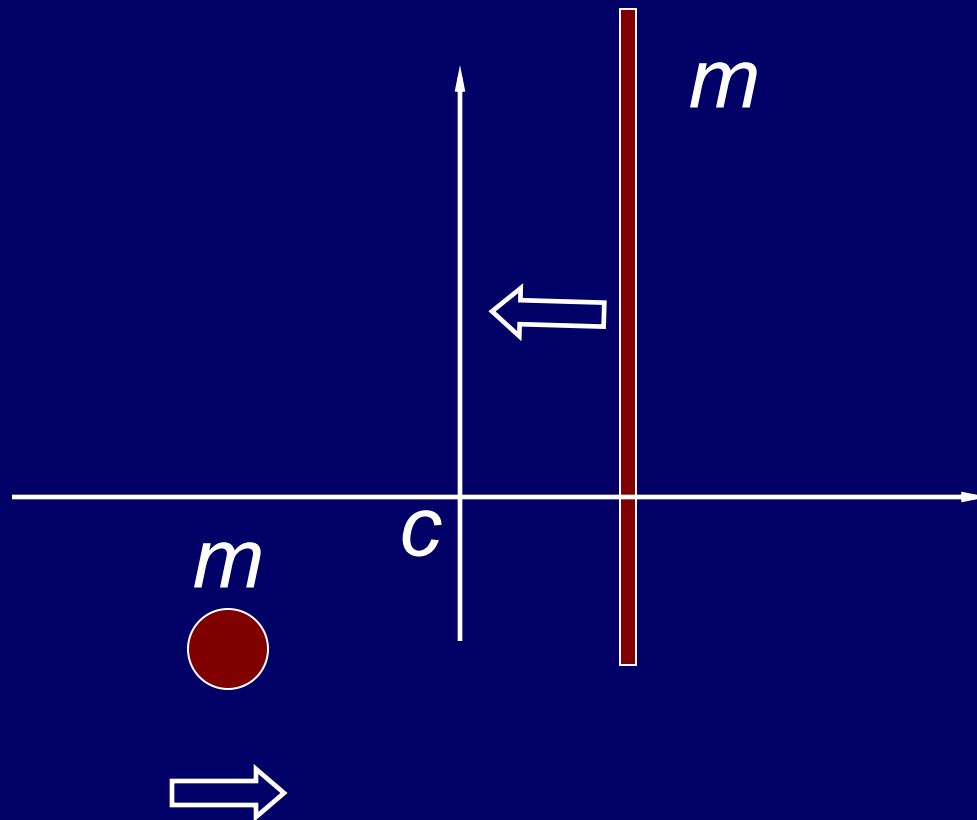
cm Frame



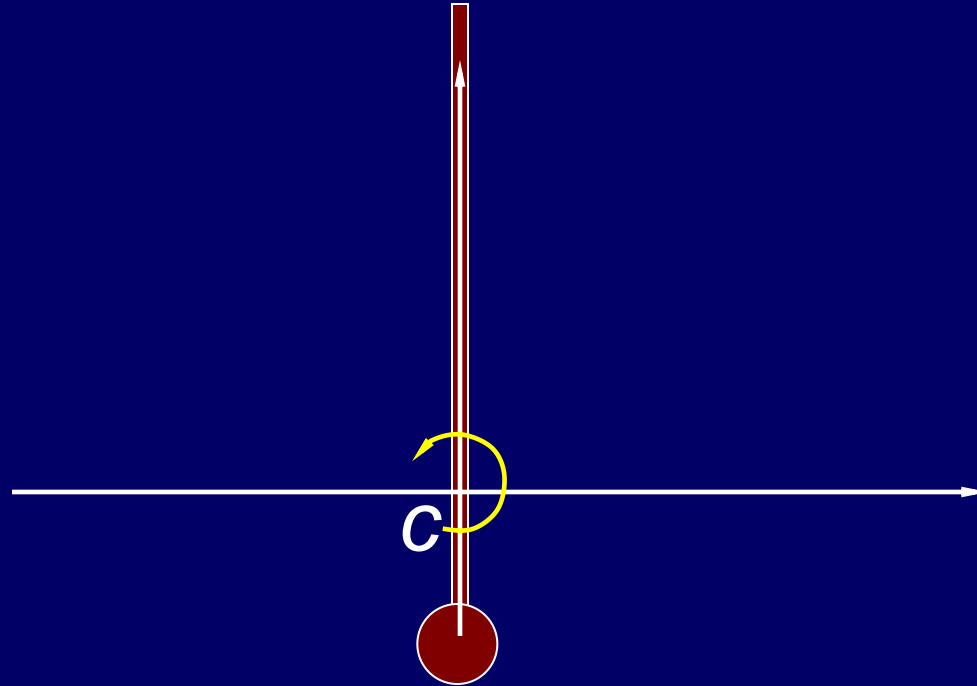
cm Frame



cm Frame



cm Frame



- The motion about **mass center**

Conservation of momentum in a system of particles

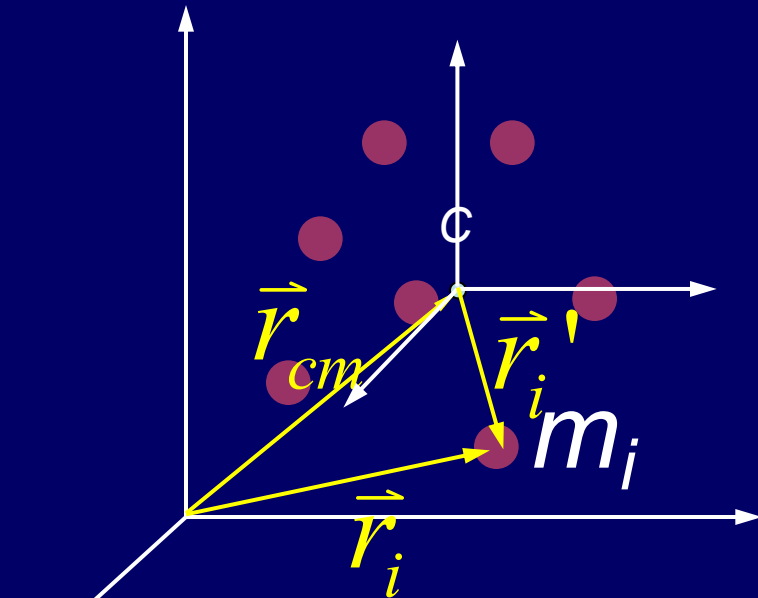
$$\vec{P} = \sum_{i=1}^N \vec{p}_i = \sum_{i=1}^N m_i \vec{v}_i = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \cdots + m_N \vec{v}_N = M \vec{v}_{cm}$$

$$\frac{d\vec{P}}{dt} = M \frac{d\vec{v}_{cm}}{dt} = M \vec{a}_{cm} = \sum \vec{F}_{ex}$$

If $\sum \vec{F}_{ext} = 0$, the total linear momentum $\vec{P}$ of the system remains unchanged. This is the law of conservation of momentum in a system of particles. It can be written as three components formula.

Lab Frame

cm Frame



In the **cm frame** the total momentum is zero

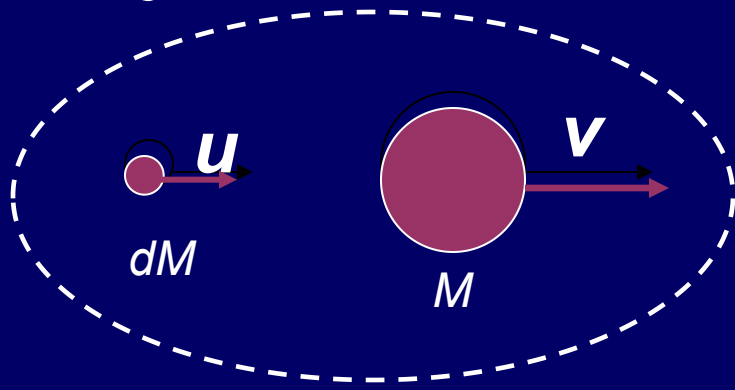
The **cm frame** called zero momentum reference frame.

$$\vec{r}'_i = \vec{r}_i - \vec{r}_{cm} \quad \vec{v}'_i = \vec{v}_i - \vec{v}_{cm} \quad \vec{a}'_i = \vec{a}_i - \vec{a}_{cm}$$

$$\vec{P}' = \sum_{i=1}^N m_i \vec{v}'_i = \sum_{i=1}^N m_i \vec{v}_i - \sum_{i=1}^N m_i \vec{v}_{cm} \quad \vec{P}' = M \vec{v}'_{cm} = 0$$

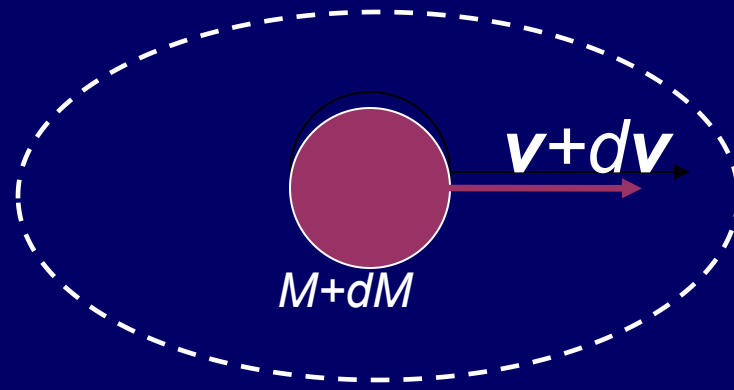
$$= M \vec{v}_{cm} - \vec{v}_{cm} \sum_{n=1}^N m_n = M \vec{v}_{cm} - \vec{v}_{cm} M = 0$$

Systems of variable mass



time t

$$\vec{P}_i = M\vec{v} + dM\vec{u}$$



time $t+dt$

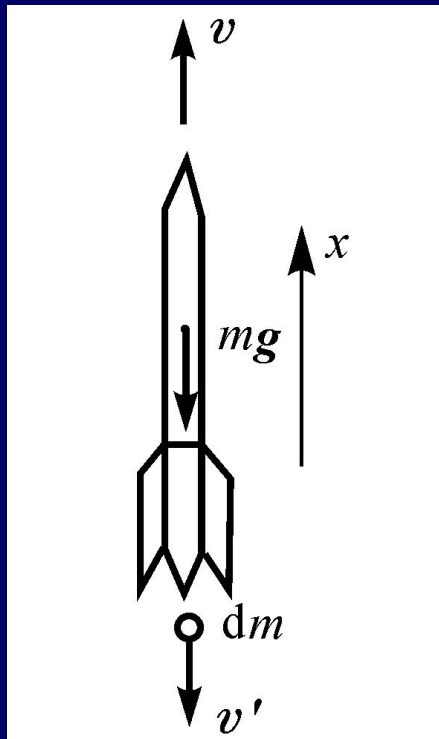
$$\vec{P}_f = (M + dM)(\vec{v} + d\vec{v})$$

$$d\vec{P} = \vec{P}_f - \vec{P}_i = M\vec{v} + M d\vec{v} + \vec{v} dM + dM d\vec{v} - M\vec{v} - \vec{u} dM$$

$$\approx M d\vec{v} - (\vec{u} - \vec{v}) dM \quad \vec{v}_{rel} = \vec{u} - \vec{v}$$

$$M \frac{d\vec{v}}{dt} = \sum \vec{F}_{ext} + \vec{v}_{rel} \frac{dM}{dt}$$

Systems of variable mass



$$\sum \vec{F}_{ext} = -mg \quad v_{rel} = -v'$$

$$dv = -gdt - v' \frac{dm}{m}$$

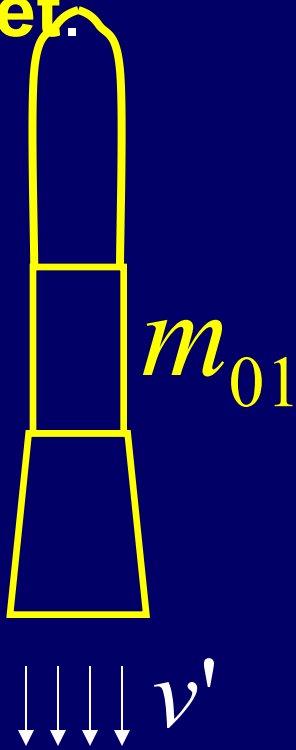
$$\int_0^v dv = -g \int_0^t dt - v' \int_{m_0}^m \frac{dm}{m}$$

If we neglect gravity

$$v = v' \ln \frac{m_0}{m}$$

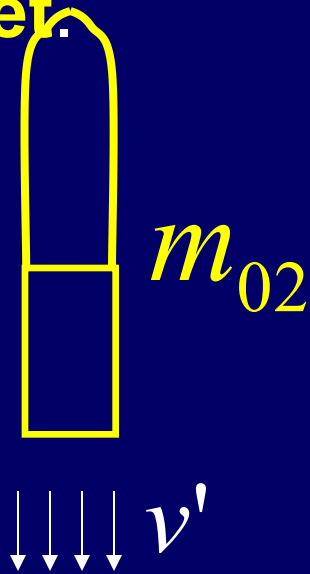
$$M \frac{d\vec{v}}{dt} = \sum \vec{F}_{ext} + \vec{v}_{rel} \frac{dM}{dt}$$

To raise the velocity of the spaceship, use **multistage rocket**.



$$v_1 - 0 = v' \ln \frac{m_{01}}{m_1}$$

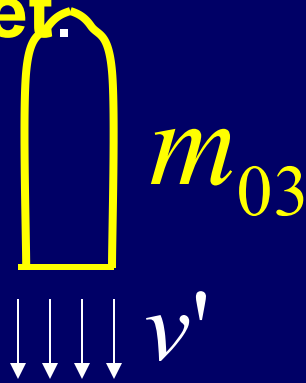
To raise the velocity of the spaceship, use **multistage rocket**.



$$v_1 - 0 = v' \ln \frac{m_{01}}{m_1}$$

$$v_2 - v_1 = v' \ln \frac{m_{02}}{m_2}$$

To raise the velocity of the spaceship, use **multistage rocket**.



$$v_1 - 0 = v' \ln \frac{m_{01}}{m_1}$$

$$v_2 - v_1 = v' \ln \frac{m_{02}}{m_2}$$

$$v_3 - v_2 = v' \ln \frac{m_{03}}{m_3}$$

$$v_3 = v' \left(\ln \frac{m_{01}}{m_1} + \ln \frac{m_{02}}{m_2} + \ln \frac{m_{03}}{m_3} \right)$$

$$\text{if } \frac{m_{01}}{m_1} = \frac{m_{02}}{m_2} = \frac{m_{03}}{m_3} = 5 \quad = v' \ln 125 = 4.8 v'$$

Sands drop from a stationary hopper at a rate of 0.134 kg/s onto a conveyor belt moving with a speed of 0.96m/s. What net force must be applied to the belt in order to keep it moving at a constant speed?



$$\frac{dM}{dt} = 0.134$$

$$\frac{d\vec{v}}{dt} = 0$$

$$\vec{v}_{rel} = -0.96$$

$$\boxed{M \frac{d\vec{v}}{dt} = \sum \vec{F}_{ext} + \vec{v}_{rel} \frac{dM}{dt}} \quad \sum \vec{F}_{ext} = -\vec{v}_{rel} \frac{dM}{dt} = 0.129 N$$

Example 1: The fuel burning rate of a rocket is $\alpha = 1.29 \times 10^3 \text{ kg/s}$. The weight of the rocket is $M_i = 2.72 \times 10^6 \text{ kg}$. (1) How large speed of the ejected gas is needed for the rocket taking off? If the ejected gas with a constant speed $u = 5.5 \times 10^4 \text{ m/s}$ lasts $T = 155 \text{ s}$, (2) how large speed of the rocket can reach? and (3) how is the rocket away from the Earth?

Answer:
$$M \frac{d\vec{v}}{dt} = \sum \vec{F}_{ext} + \vec{v}_{rel} \frac{dM}{dt}$$

$$1) \quad M_i g = v dM / dt \quad v = M_i g / \alpha = 2.07 \times 10^4 \text{ m/s}$$

$$2) \quad M \frac{dv}{dt} = -Mg - u \frac{dM}{dt} \quad v = - \int_{M_i}^{M_f} \frac{u}{M} dM - \int_0^T g dt = u \ln \frac{M_i}{M_f} - gT$$

$$M_f = M_i - \alpha T \quad v = 2.68 \times 10^3 \text{ m/s}$$

$$3) \quad h = \int_0^T v dt = \int_0^T \left[u \ln \frac{M_i}{M_i - \alpha t} - gt \right] dt = 172 \text{ km}$$

CHAP 7

Exercises

P155:11, 13, 14, 17, 18

Problems

P157:3,10